Remaining audit items

25 groups: 17 dissertation/shared items and 8 companion-paper items. Status labels separate existing-record checks, proposed corrections, and optional editorial choices. Yellow marks what remains to decide or address.

Source lines are one-based and remapped to the current working files, including the combined Chapter 6 Methods and Verification. The full Markdown carries clickable file paths and detail; the separate crosswalk accounts for resolved/removed IDs. The original 83-page audit is unchanged.

Existing results and published Chapter 3 remain protected. This list does not assign FVU reanalysis, AI-rating methods/reclassification, new optimization, simulations/refits, the waived individual-vastus example, or new fixes for retained author objections.

Dissertation and shared material

D01 Document the historical gait score behind each result family

Existing-record verification

The historical objective is not operationally identified.

Action / decision: Match each saved result family to its archived scorer/version. Define its equation or terms and distinguish the no-cycle sentinel. Preserve recorded scores.

Location: §D.3 Tuning and Supported Evaluations; §D.3.2 Direct Kinematics Overlay

Source: 95-AppendixD.tex:127, 157

Find: “zero to perfect agreement”

Original IDs: SNS-S01

D02 Define duty and knee-minimum denominators for the zero-cycle case

Existing-definition verification

The caption is cycle-conditioned, but the left leg has zero cycles and a duty of 0.985.

Action / decision: Check the saved duty and knee-minimum definitions; label the actual denominator, including the zero-cycle leg. Keep reported values.

Location: §D.4.1 MuJoCo Network and Ground Evaluation

Source: 95-AppendixD.tex:222, 229, 231

Find: “computed over detected cycles”

Original IDs: SNS-S02

D03 Identify the earlier spiking knee comparison as a startup configuration

Clarification proposal

Action / decision: Identify the earlier startup configuration, use final-window labels where settling was not demonstrated, and state its angle convention and relationship to the later broadened-ramp transient test.

Location: §D.4.2 Simulink Demonstrations; §6.4.3 SNS-Simscape Block Library and Editor

Source: 95-AppendixD.tex:251; 60-sns.tex:141

Find: “settle statistics”

Original IDs: SNS-S03, O-A08

D04 Provide one compact model/controller/support map

Presentation decision

Action / decision: Consider one appendix map: model/mechanics, controller, saved ID, actuator law, support, outcome, and figure/table. Replace repeated inventory prose and identify the production configuration precisely.

Location: §6.4.2 Gait2392-Based Models; §D.3.1 Reference-Library Comparison; §D.5.4 CAD-Derived Robot Integration
Source: 60-sns.tex:122; 95-AppendixD.tex:133, 358
Find: “production configuration”
Original IDs: SNS-S08, LATER-37

D05 Enlarge the inspectable SNS diagram details

Figure-polish decision

Action / decision: Give the neuron signal-flow/ports a readable crop and enlarge the knee inset inside the existing composite. Retain the twelve-block overview and all six feedback connections. Also replace the attached C in the knee-demo citation with bold parenthesized (C), preserving its figure label.

Location: §6.4.3 SNS-Simscape Block Library and Editor
Source: 60-sns.tex:141, 149, 156
Find: “sns_library_implementation_current.pdf”
Also find: “Figure~\ref{fig:sns_reflex_current}C”
Original IDs: SNS-S05

D06 Show the contact-phase result in the W2L comparison itself

Figure-annotation decision

Action / decision: Annotate the saved median phase and 0.35-0.65 acceptance interval in each figure column. Use matching comparison-axis limits where appropriate; retain every trace.

Location: §6.4.1 Li-Deng/W2L Walker and Controller
Source: 60-sns.tex:83, 95
Find: “0.498 to 0.223 cycles”
Original IDs: SNS-S10

D07 Separate optional software acceptance bookkeeping from appendix evidence

Optional presentation decision

Action / decision: Retain the headless-only scope and identity-test outcome. Consider keeping GUI acceptance status and full 17-digit strings in the archived report rather than the scientific appendix.

Location: §B.4.3 Verification Gates of the MuJoCo Mirror; §B.4.4 AnimatLab Conversion Variants
Source: 93-AppendixB.tex:152, 200
Find: “Non-spiking bit-identity”
Original IDs: APP-A5

D08 Use reader-facing status terms in Appendix D/B

Wording proposal

Action / decision: Replace debug terms with explicit outcomes, separate the non-finite/fall fields, and clarify zero external neural drive. Keep correct censuses, failed outcomes, and assistance qualifications.

Location: §D.4.1 MuJoCo Network and Ground Evaluation; §D.5 Supported BPA-Robot Evaluations; §D.5.4 CAD-Derived Robot Integration; §B.4.4 AnimatLab Conversion Variants

Source: 95-AppendixD.tex:233, 304, 360; 93-AppendixB.tex:177

Find: "Non-finite states / fell"

Original IDs: SNS-S11

D09 Improve the remaining route-geometry and transmission figure details

Figure/readability decision

Action / decision: Supply the main pose frame/axis names/units and uncrowd contact labels. Enlarge Appendix C route/transmission details; crop the bone view or omit it only if it adds no unique geometry. Direct-route and cam views are distinct configurations.

Location: §4.3.3 Redesigned Routes for the Biomimetic Robot Model; §C.6 Redesigned Extensor Route Geometry; §C.8 Supporting Route Searches and Transmission Sweeps

Source: 30-torque.tex:715; 94-AppendixC.tex:325, 333, 406, 413

Find: "Cam-extensor geometry at"

Original IDs: T14, APP-A3

D10 Remove repeated full-corpus inventories from Appendix E

Selective compression decision

Action / decision: Replace repeated full-set inventory/graph lists with Chapter 5 references; retain the unique 561-paper maintenance counts. A coverage table is optional. Consider moving the III/IV convergence explanation into the pathway-role column.

Location: §5.2 Results: Application and Corpus Coverage; §E.1 Record Sets and Coverage; §E.2 Afferent Classes and Linked Pathways

Source: 50-database.tex:31; 96-AppendixE.tex:10, 14, 33

Find: "943 entries"

Original IDs: DB-S5, SADB_AUDIT-Appendix relevance (unnumbered P3)

D11 Identify the retrievable SADb snapshot

Availability/provenance decision

Action / decision: Identify the existing retrievable app/repository and the archived snapshot behind the 1,393-record counts and captures. Add a verified version/date if available; do not invent a release or hosting.

Location: Chapter 5 opening; §5.1 Curation Methods; Appendix E opening

Source: 50-database.tex:8, 25; 96-AppendixE.tex:4

Find: "SADb Explorer"

Original IDs: DB-S6

D12 Add only the useful local terminology reminders

Optional gloss/title decision

Action / decision: Consider a local l620 reminder, Hunt-model citation, a reeving heading/definition separating BPA count from cable parts/sheaves, and a brief OpenAlex gloss. The n/G pulley equations already define the transmission.

Location: §4.1.3 Hybrid Torque Calculation Method; §4.2 Results; §4.3.5 Mono- and Bi-Pulley Tendon Routing; Chapter 5 opening

Source: 30-torque.tex:83, 472, 764; 50-database.tex:8

Find: "l_{620}"

Original IDs: LATER-34, LATER-35, LATER-36, LATER-42

D13 Check the existing reuse records for three reproduced images

Source-specific verification

Action / decision: Locate the actual source and existing reuse basis for Kengoro, Rybak 2006 Figure 1, and Shevtsova Figure 5.8. Apply each required credit; do not insert blanket CC BY wording or borrow the 2015 review license.

Location: §2.1 Biomimetic and Musculoskeletal Robots; §2.5 Neural Control of Locomotion: Central Pattern Generators

Source: 15-background.tex:19, 169, 181

Find: "Kengoro human-mimetic"

Original IDs: C2-O15, C2-O16, C2-O17, LATER-56, LATER-57, LATER-58

Sources: Kengoro primary record, Rybak 2006 source, Shevtsova chapter

D14 Confirm the torque manuscript author list and title

Human metadata decision

The dissertation note lists seven torque authors; the paper lists four.

Action / decision: Have the authors confirm the intended list, contribution statement, title, and status; then align the dissertation note, reference, and paper metadata.

Location: Chapter 4 opening; Paper front matter; References

Source: 30-torque.tex:5; Main.tex:48, 69; thesis.bib:1113

Find: "unpublished manuscript in preparation"

Original IDs: T11

D15 Choose the scope of prior-publication/presentation notes

Author decision; no insertion authorized

Action / decision: Choose one concise Chapter 5 presentation credit or keep the existing opener; confirm the 2023 link. Add a scoped Chapter 4 prior-publication/contribution note only if earlier co-authored material is incorporated. A citation or poster alone does not establish whole-chapter republication.

Location: Chapter 5 opening; §4.1.5 Actuator Placement and Route Redesign; References

Source: 50-database.tex:4; 30-torque.tex:342; thesis.bib:1132

Find: "presented at Neuroscience 2023 and 2024"

Original IDs: PUBLICATION_NOTE_RECOMMENDATION

Sources: PSU ETD handbook, March 2026

D16 Choose limited compression of the mechanical methods

Optional compression decision

Action / decision: Optionally shorten repeated compliance substitutions and legacy search implementation details. Preserve the bracket development, core mechanics, master's/Morrow attribution, and distinct mesh objective. Keep the author-retained route-builder chronology.

Location: §4.1.4 / Length Offset and System Stiffness (unnumbered subheading); Paper §2.6 Improved Torque Model: Length Offset and Bracket Compliance; §4.1.5 Actuator Placement and Route Redesign

Source: 30-torque.tex:91, 344, 346; Main.tex:197

Find: "Length Offset and System Stiffness"

Original IDs: T13, LATER-46

D17 Separate the two meanings of rho in the contact-release description

Notation-fix proposal

One paragraph assigns rho to both curvature and a fixed relaxation.

Action / decision: Use a separate symbol for fixed clearance relaxation in Appendix C and the paper supplement; retain rho(theta) for curvature and keep the 3 mm value.

Location: §C.7 Contact and Release Angles of the Wrapped Route; Supplement §1.5 Contact and Release Angles of the Wrapped Route

Source: 94-AppendixC.tex:377; frontiers_SupplementaryMaterial.tex:251

Find: "\rho = \qty{3}{\mm}"

Original IDs: OP07, LATER-49

Companion paper and supplement

P01 Update the published force-model reference

Bibliographic correction proposal

Action / decision: Update the stale record to the published Actuators 15(5):230 metadata and DOI 10.3390/act15050230. Preserve its cited key and use byte-safe bibliography handling.

Location: References; Paper §2.2 Actuator Force Model

Source: BolenFrontiers22.bib:1085; Main.tex:112

Find: "@unpublished{bolen_2026"

Original IDs: T05

Sources: Published force paper

P02 Check the remaining small pinned/supplementary plot labels

Figure-polish decision

The unchanged audited assets had legends near 6.3 and 5.3 pt.

Action / decision: Improve final-size labels in the pinned-extensor and supplementary direct-route figures to the standing 10-pt target. Preserve all traces. This excludes the already repaired physical pair and extensor legend.

Location: Paper §3.2 Extensor on Pin Joint; Supplement §1.3 Figures

Source: Main.tex:443; frontiers_SupplementaryMaterial.tex:173

Find: "fig:ExtPin_group"

Original IDs: T07

P03 Decide whether the journal should place hardware evidence in Results

Optional paper-structure decision

Action / decision: For journal organization, consider moving hardware capacity evidence into Results, leaving interpretation in Discussion. Keep the approved result-first sequence and pulley placement; this does not move dissertation §4.3.4.

Location: Paper §4.3.2 Physical Validation Against Human Torque; Paper §4.4 Whole-Leg Capability Against Human Torque Surfaces
Source: Main.tex:600, 623
Find: “Physical Validation Against Human Torque”
Original IDs: T08

P04 Replace the remaining identification settings wall with a compact specification

Compression proposal

Action / decision: Consider a two-row stage/settings table and short explanation. Preserve all training/validation memberships, bounds, optimizer settings, and the shared stiffness constraint. Remove duplicated prose, with no new optimization or FVU analysis.

Location: Paper §2.8 Identification Procedure
Source: Main.tex:382
Find: “The identification proceeded in two stages.”
Original IDs: T09

P05 Represent both hardware routes in the paper abstract

Framing/optional compression proposal

Action / decision: Include the existing flexor as well as extensor capacity result in the abstract, scoped to tested configurations/points. Optionally trim repeated Introduction/Overview work lists. The 27-site segue is already fixed.

Location: Paper front matter; Paper §4.3.2 Physical Validation Against Human Torque; Paper §2.1 Overview
Source: Main.tex:81, 602, 108
Find: “first hardware torque measurements of the redesigned extensor route”
Original IDs: T12, OP05, LATER-54

P06 Add the remaining pinned-bracket inventory callout

Small callout proposal

Action / decision: Add one explicit numbered reference to the pinned-bracket table in its existing introduction or the relevant main Methods sentence. Other retained coordinate/cam items now have useful calls.

Location: Supplement §1.2 BPA routing and via points
Source: frontiers_SupplementaryMaterial.tex:80, 84
Find: “Bracket reference points of the pinned-knee BPA routes”
Original IDs: T15, OP14

P07 Make first table mentions follow table order

Callout-order proposal

The adopted-parameter table is mentioned before the first two main tables.

Action / decision: Defer/remove the two premature adopted-parameter table calls; retain the values and introduce the table naturally in §4.2. Recheck first-mention order after the build, using labels rather than manual numbering.

Location: Paper §3.1 Flexor on Pin Joint; Paper §3.2 Extensor on Pin Joint; Paper §4.2 Identified Design Parameters
Source: Main.tex:411, 437, 551
Find: “adopted design parameters”
Original IDs: OP13, LATER-52
Sources: Frontiers figure/table guidance

P08 Optional bibliography source cleanup

Optional source-hygiene decision

Action / decision: Optionally remove the unused supplement bibliography invocation while it has no citations, and consolidate the uncited Hunt duplicate while retaining the cited key. Neither is a current duplicate/mid-document printed-reference defect.

Location: Supplement §1.3 Figures; References
Source: frontiers_SupplementaryMaterial.tex:177; BolenFrontiers22.bib:369, 1020
Find: “\bibliography{BolenFrontiers22}”
Original IDs: OP11, OP15, LATER-51, LATER-53